

KUBERNETES TASK 9

Task 1: Execute all the steps shown in video.

Environment: Minikube (Docker driver) on Windows, using Git Bash or PowerShell.

Step 1: Start Minikube

```
minikube start --driver=docker
```

Wait for the command to finish. kubectl is now configured to use the "minikube" cluster.

```
HP@DESKTOP-5IFH5B0 MINGW64 ~/Downloads
$ minikube start --driver=docker
* minikube v1.39.0 on Microsoft Windows 11 Pro 26H2
* Using the docker driver based on existing profile
* Starting "minikube" primary control-plane node in "minikube" cluster
* Pulling base image v0.0.51 ...
* Preparing Kubernetes v1.37.0 on containerd 2.3.4 ...
* Verifying Kubernetes components...
  - Using image docker.io/kubernetes/dashboard:v2.7.0
  - Using image docker.io/kubernetes/metrics-scraper:v1.0.8
  - Using image gcr.io/k8s-minikube/storage-provisioner:v5
* Some dashboard features require the metrics-server addon. To enable all features please run:

    minikube addons enable metrics-server

* Enabled addons: storage-provisioner, dashboard, default-storageclass
* Done! kubectl is now configured to use "minikube" cluster and "default" namespace by default
```

```
minikube status
```

Expected: host, kubelet and apiserver are Running; kubeconfig is Configured.

```
PS C:\Users\HP\Desktop> minikube status
minikube
type: Control Plane
host: Running
kubelet: Running
apiserver: Running
kubeconfig: Configured
```

```
kubectl get nodes
```

Expected: the minikube node is in Ready state with the control-plane role.

```
PS C:\Users\HP\Desktop> kubectl get nodes
NAME          STATUS    ROLES          AGE    VERSION
minikube      Ready    control-plane  10d    v1.37.0
PS C:\Users\HP\Desktop> █
```

```
PS C:\Users\HP\Desktop> kubectl get pods -A
NAMESPACE      NAME                                                    READY   STATUS    RESTARTS   AGE
kube-system    coredns-559f6c778d-tg5s2                             1/1     Running   1 (12m ago)  10d
kube-system    etcd-minikube                                           1/1     Running   1 (12m ago)  10d
kube-system    kindnet-5w9cd                                           1/1     Running   1 (12m ago)  10d
kube-system    kube-apiserver-minikube                               1/1     Running   1 (12m ago)  10d
kube-system    kube-controller-manager-minikube                     1/1     Running   1 (12m ago)  10d
kube-system    kube-proxy-xgg2h                                        1/1     Running   1 (12m ago)  10d
kube-system    kube-scheduler-minikube                               1/1     Running   1 (12m ago)  10d
kube-system    storage-provisioner                                    1/1     Running   2 (12m ago)  10d
kubernetes-dashboard dashboard-metrics-scraper-69cbd87b98-lkcws             1/1     Running   1 (12m ago)  10d
kubernetes-dashboard kubernetes-dashboard-6df74558bd-l6xrq               1/1     Running   2 (11m ago)  10d
PS C:\Users\HP\Desktop> █
```

Step 2: Create the NGINX Deployment

```
nano deployment.yaml
apiVersion: apps/v1
kind: Deployment
metadata:
  name: nginx-deployment
spec:
  replicas: 1
  selector:
    matchLabels:
      app: nginx
  template:
    metadata:
      labels:
        app: nginx
    spec:
      containers:
        - name: nginx
          image: nginx
          ports:
            - containerPort: 80
```

```
MINGW64:/c/Users/HP/Desktop/k8s-ingress-testing
GNU nano 6.3 deployment.yaml
apiVersion: apps/v1
kind: Deployment
metadata:
  name: nginx-deployment
spec:
  replicas: 1
  selector:
    matchLabels:
      app: nginx
  template:
    metadata:
      labels:
        app: nginx
    spec:
      containers:
      - name: nginx
        image: nginx
        ports:
        - containerPort: 80
```

```
kubectl apply -f deployment.yaml
kubectl get deployment
kubectl get pods -o wide
```

Expected: the nginx-deployment pod shows 1/1 Running.

```
HP@DESKTOP-SIFH580 MINGW64 ~/Desktop/k8s-ingress-testing (main)
$ kubectl apply -f deployment.yaml
deployment.apps/nginx-deployment created

HP@DESKTOP-SIFH580 MINGW64 ~/Desktop/k8s-ingress-testing (main)
$ kubectl get deployment
NAME          READY   UP-TO-DATE   AVAILABLE   AGE
nginx-deployment 0/1      1            0           10s

HP@DESKTOP-SIFH580 MINGW64 ~/Desktop/k8s-ingress-testing (main)
$ kubectl get pods -o wide
NAME                                READY   STATUS    RESTARTS   AGE   IP            NODE     NOMINATED NODE   READINESS GATES
nginx-deployment-8648945fd5-rblfz  1/1     Running   0           20s   10.244.0.5    minikube <none>         <none>
```

Step 3: Create the NodePort Service

```
nano service.yaml

apiVersion: v1
kind: Service
metadata:
  name: nginx-service
spec:
  type: NodePort
  selector:
    app: nginx
  ports:
    - port: 80
      targetPort: 80
      nodePort: 30001
```

```
MINGW64:/c/Users/HP/Desktop/k8s-ingress-testing
GNU nano 6.3 service.yaml
apiVersion: v1
kind: Service
metadata:
  name: nginx-service
spec:
  type: NodePort
  selector:
    app: nginx
  ports:
  - port: 80
    targetPort: 80
    nodePort: 30001
```

```
kubectl apply -f service.yaml
kubectl get svc
kubectl get endpoints nginx-service
```

Expected: nginx-service of type NodePort mapped as 80:30001/TCP, and the endpoint lists the NGINX pod IP.

```
HP@DESKTOP-5IFH5BO MINGW64 ~/Desktop/k8s-ingress-testing (main)
$ kubectl apply -f service.yaml
service/nginx-service created

HP@DESKTOP-5IFH5BO MINGW64 ~/Desktop/k8s-ingress-testing (main)
$ kubectl get svc
NAME                TYPE          CLUSTER-IP    EXTERNAL-IP    PORT(S)          AGE
kubernetes          ClusterIP     10.96.0.1     <none>         443/TCP          11d
nginx-service       NodePort      10.107.5.80   <none>         80:30001/TCP     11s

HP@DESKTOP-5IFH5BO MINGW64 ~/Desktop/k8s-ingress-testing (main)
$ kubectl get endpoints nginx-service
Warning: v1 Endpoints is deprecated in v1.33+; use discovery.k8s.io/v1 EndpointSlice
NAME                ENDPOINTS          AGE
nginx-service       10.244.0.5:80      25s

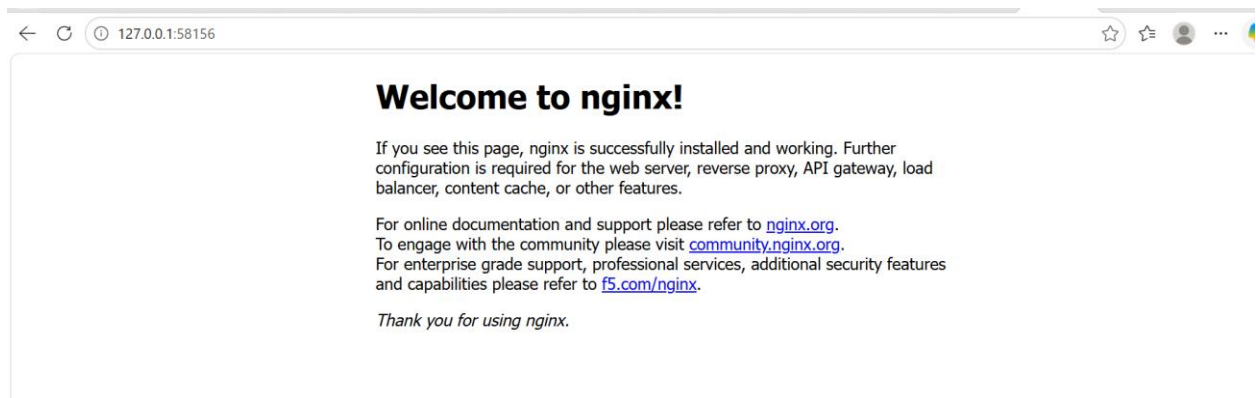
HP@DESKTOP-5IFH5BO MINGW64 ~/Desktop/k8s-ingress-testing (main)
$
```

Step 4: Test the NodePort Service

```
minikube service nginx-service --url
```

```
HP@DESKTOP-5IFH5BO MINGW64 ~/Desktop/k8s-ingress-testing (main)
$ minikube service nginx-service --url
http://127.0.0.1:58156
! Because you are using a Docker driver on windows, the terminal needs to be open to run it.
```

Open the returned URL (for example, <http://192.168.x.x:30001>) in a browser. The NGINX welcome page should be displayed.



Step 5: Enable the Ingress Controller

```
minikube addons enable ingress
```

```
HP@DESKTOP-5IFH580 MINGW64 ~/Desktop/k8s-ingress-testing (main)
$ minikube addons enable ingress
* ingress is an addon maintained by Kubernetes. For any concerns contact minikube on GitHub.
You can view the list of minikube maintainers at: https://github.com/kubernetes/minikube/blob/master/OWNERS
* After the addon is enabled, please run "minikube tunnel" and your ingress resources would be available at "127.0.0.1"
  - Using image registry.k8s.io/ingress-nginx/controller:v1.15.1
  - Using image registry.k8s.io/ingress-nginx/kube-webhook-certgen:v1.6.9
  - Using image registry.k8s.io/ingress-nginx/kube-webhook-certgen:v1.6.9
* Verifying ingress addon...
* The 'ingress' addon is enabled
```

```
kubectl get pods -n ingress-nginx
kubectl get pods -A | grep ingress
```

Expected: the ingress-nginx controller pod is Running.

```
HP@DESKTOP-5IFH580 MINGW64 ~/Desktop/k8s-ingress-testing (main)
$ kubectl get pods -n ingress-nginx
NAME                                READY   STATUS    RESTARTS   AGE
ingress-nginx-admission-create-kwtz6 0/1     Completed 0           63s
ingress-nginx-admission-patch-lnbn4   0/1     Completed 0           63s
ingress-nginx-controller-d7cd8c989-kg87s 1/1     Running   0           63s

HP@DESKTOP-5IFH580 MINGW64 ~/Desktop/k8s-ingress-testing (main)
$ kubectl get pods -A | grep ingress
ingress-nginx      ingress-nginx-admission-create-kwtz6      0/1     Completed    0           82s
ingress-nginx      ingress-nginx-admission-patch-lnbn4      0/1     Completed    0           82s
ingress-nginx      ingress-nginx-controller-d7cd8c989-kg87s 1/1     Running      0           82s

HP@DESKTOP-5IFH580 MINGW64 ~/Desktop/k8s-ingress-testing (main)
$
```

Step 6: Create the Ingress Resource

```
nano ingress.yaml

apiVersion: networking.k8s.io/v1
kind: Ingress
metadata:
  name: nginx-ingress
spec:
  ingressClassName: nginx
  rules:
  - host: nginx.local
    http:
```

```
paths:
- path: /
  pathType: Prefix
  backend:
    service:
      name: nginx-service
      port:
        number: 80
```

MINGW64:/c/Users/HP/Desktop/k8s-ingress-testing

```
GNU nano 6.3 ingress.yaml
apiVersion: networking.k8s.io/v1
kind: Ingress
metadata:
  name: nginx-ingress
spec:
  ingressClassName: nginx
  rules:
  - host: nginx.local
    http:
      paths:
      - path: /
        pathType: Prefix
        backend:
          service:
            name: nginx-service
            port:
              number: 80
```

```
kubectl apply -f ingress.yaml
kubectl get ingress
```

The ADDRESS column may be empty at first. Wait a short while and run kubectl get ingress again.

```
HP@DESKTOP-5IFH5BO MINGW64 ~/Desktop/k8s-ingress-testing (main)
$ kubectl apply -f ingress.yaml
ingress.networking.k8s.io/nginx-ingress created

HP@DESKTOP-5IFH5BO MINGW64 ~/Desktop/k8s-ingress-testing (main)
$ kubectl get ingress
NAME          CLASS  HOSTS          ADDRESS  PORTS  AGE
nginx-ingress  nginx  nginx.local    80       14s

HP@DESKTOP-5IFH5BO MINGW64 ~/Desktop/k8s-ingress-testing (main)
$
```

Step 7: Get the Minikube IP

```
minikube ip
```

```
HP@DESKTOP-5IFH5BO MINGW64 ~/Desktop/k8s-ingress-testing (main)
$ minikube ip
192.168.49.2

HP@DESKTOP-5IFH5BO MINGW64 ~/Desktop/k8s-ingress-testing (main)
$ |
```

Step 8: Configure the Windows Hosts File

Open **Notepad as Administrator**, then open the hosts file:

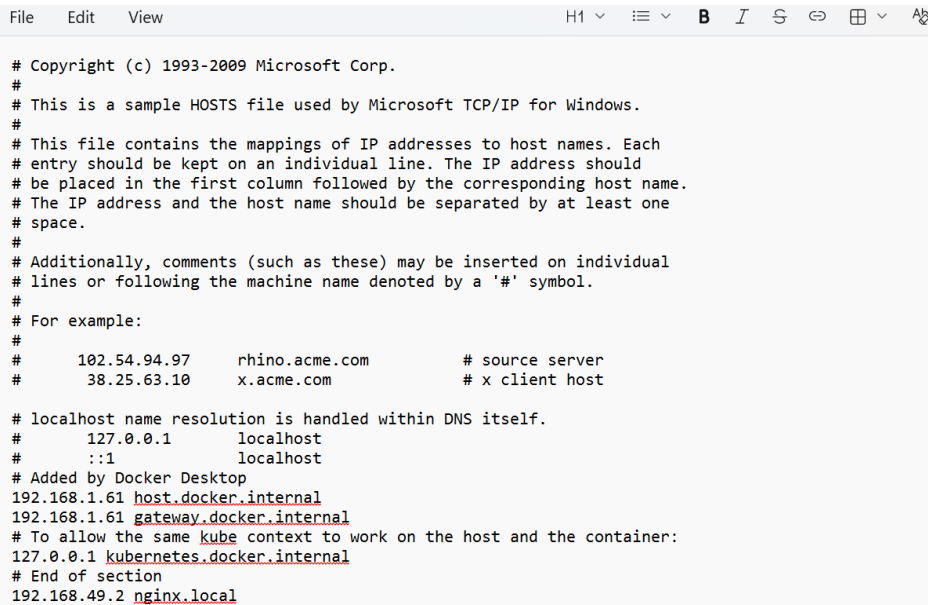
```
C:\Windows\System32\drivers\etc\hosts
```

Add the following entry (replace the IP with the output of minikube ip) and save the file:

```
<MINIKUBE-IP> nginx.local
```

Example:

```
192.168.49.2 nginx.local
```



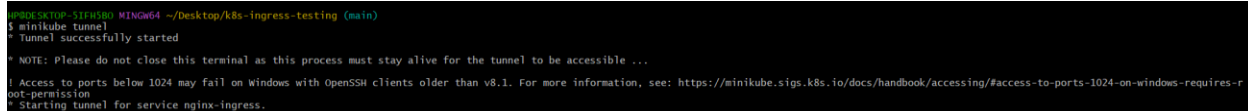
```
File Edit View H1  B I S  A
# Copyright (c) 1993-2009 Microsoft Corp.
#
# This is a sample HOSTS file used by Microsoft TCP/IP for Windows.
#
# This file contains the mappings of IP addresses to host names. Each
# entry should be kept on an individual line. The IP address should
# be placed in the first column followed by the corresponding host name.
# The IP address and the host name should be separated by at least one
# space.
#
# Additionally, comments (such as these) may be inserted on individual
# lines or following the machine name denoted by a '#' symbol.
#
# For example:
#
#       102.54.94.97     rhino.acme.com   # source server
#       38.25.63.10     x.acme.com      # x client host
#
# localhost name resolution is handled within DNS itself.
#       127.0.0.1       localhost
#       ::1             localhost
# Added by Docker Desktop
192.168.1.61 host.docker.internal
192.168.1.61 gateway.docker.internal
# To allow the same kube context to work on the host and the container:
127.0.0.1 kubernetes.docker.internal
# End of section
192.168.49.2 nginx.local
```

Step 9: Start the Minikube Tunnel

Open a **second terminal** and run:

```
minikube tunnel
```

Keep this terminal open. Closing the tunnel makes the application inaccessible.



```
minikube@minikube: ~/Desktop/k8s-ingress-testing (main)
$ minikube tunnel
" Tunnel successfully started

* NOTE: Please do not close this terminal as this process must stay alive for the tunnel to be accessible ...
! Access to ports below 1024 may fail on Windows with OpenSSH clients older than v8.1. For more information, see: https://minikube.sigs.k8s.io/docs/handbook/accessing/#access-to-ports-1024-on-windows-requires-r
out-permission
* Starting tunnel for service nginx-ingress.
```

Open the application in a browser:

```
http://nginx.local
```

Expected: the "Welcome to nginx!" page.

Welcome to nginx!

If you see this page, nginx is successfully installed and working. Further configuration is required for the web server, reverse proxy, API gateway, load balancer, content cache, or other features.

For online documentation and support please refer to nginx.org.

To engage with the community please visit community.nginx.org.

For enterprise grade support, professional services, additional security features and capabilities please refer to f5.com/nginx.

Thank you for using nginx.

Task 2: Upgrade Kubernetes from 1.26 to 1.28

Cluster: k8s-master (control plane), k8s-worker1, k8s-worker2.

Upgrade path:

1. Upgrade from 1.26 to 1.27.
2. Upgrade from 1.27 to 1.28.

Minor versions cannot be skipped, so the upgrade must not go directly from 1.26 to 1.28. Upgrade the control plane first, then the workers one at a time, draining each node before upgrading its kubelet.

Phase 1: Verify the Current Cluster

Run on the control plane:

```
kubectl get nodes -o wide
```

Expected: all three nodes at v1.26.x.

```
root@k8s-master:~# kubectl get nodes -o wide
NAME          STATUS    ROLES    AGE   VERSION   INTERNAL-IP   EXTERNAL-IP   OS-IMAGE             KERNEL-VERSION   CONTAINER-RUNTIME
k8s-master    Ready     control-plane  10d   v1.29.15   172.31.21.13   <none>         Ubuntu 26.04.1 LTS   7.0.0-1013-aws   containerd://2.2.2
k8s-worker1   Ready     <none>      10d   v1.29.15   172.31.19.119  <none>         Ubuntu 26.04.1 LTS   7.0.0-1013-aws   containerd://2.2.2
k8s-worker2   Ready     <none>      10d   v1.29.15   172.31.22.16   <none>         Ubuntu 26.04.1 LTS   7.0.0-1013-aws   containerd://2.2.2
root@k8s-master:~#
```

```
kubeadm version
kubelet --version
kubectl version --client
```

```
root@k8s-master:~# kubeadm version
kubeadm version: &version.Info{Major:"1", Minor:"29", GitVersion:"v1.29.15", GitCommit:"0d0f172cdf9fd42d6fdee3467374b58d3e168df0", GitTreeState:"clean", BuildDate:"2025-03-11T17:46:36Z", GoVersion:"go1.23.6", Compiler:"gc", Platform:"linux/amd64"}
root@k8s-master:~# kubelet --version
Kubernetes v1.29.15
root@k8s-master:~# kubectl version --client
Client Version: v1.29.15
Kustomize Version: v5.0.4-0.20230601165947-6ce0bf390ce3
root@k8s-master:~#
```

Phase 2: Check Cluster Health

```
kubectl get nodes
```

Expected: all nodes are Ready.

```
root@k8s-master:~# kubectl get nodes
NAME          STATUS    ROLES    AGE   VERSION
k8s-master    Ready     control-plane  10d   v1.29.15
k8s-worker1   Ready     <none>      10d   v1.29.15
k8s-worker2   Ready     <none>      10d   v1.29.15
root@k8s-master:~#
```

```
kubectl get pods -A
```

Confirm that the system pods are Running: kube-apiserver, kube-controller-manager, kube-scheduler, etcd, coredns, kube-proxy and the CNI pods.

```

root@k8s-master:~# kubectl get pods -A

```

NAMESPACE	NAME	READY	STATUS	RESTARTS	AGE
default	efs-pod1	0/1	CreateContainerError	1 (159m ago)	162m
default	efs-pod2	0/1	CreateContainerError	13 (156m ago)	3h58m
default	liveness-test	0/1	CrashLoopBackOff	66 (4m2s ago)	3h8m
default	probe-pod	0/1	CrashLoopBackOff	40 (2m18s ago)	3h28m
default	readiness-test	0/1	CrashLoopBackOff	43 (2m5s ago)	3h10m
default	test-pod	1/1	Running	32 (119s ago)	175m
dev-environment	pod1	1/1	Running	223 (7m12s ago)	2d3h
kube-system	calico-kube-controllers-8d76c5f9b-c2j8s	1/1	Running	7 (7h37m ago)	10d
kube-system	calico-node-84qk4	1/1	Running	501 (7m30s ago)	10d
kube-system	calico-node-l9xzc	0/1	CrashLoopBackOff	494 (31s ago)	10d
kube-system	calico-node-scmjx	1/1	Running	7 (7h37m ago)	10d
kube-system	coredns-76f75df574-clw82	1/1	Running	7 (7h37m ago)	10d
kube-system	coredns-76f75df574-m964x	1/1	Running	7 (7h37m ago)	10d
kube-system	ebs-csi-controller-65988b59b-54fmb	5/5	Running	486 (5m51s ago)	24h
kube-system	ebs-csi-controller-65988b59b-79spf	0/5	CrashLoopBackOff	178 (3m6s ago)	177m
kube-system	ebs-csi-node-jfswm	3/3	Running	3 (7h37m ago)	24h
kube-system	ebs-csi-node-rzm2q	0/3	CrashLoopBackOff	285 (2m ago)	24h
kube-system	ebs-csi-node-x7mc5	0/3	CrashLoopBackOff	270 (62s ago)	24h
kube-system	efs-csi-controller-56796ddf54-dqt5s	0/3	CrashLoopBackOff	210 (4m2s ago)	6h42m
kube-system	efs-csi-controller-56796ddf54-t9pm2	3/3	Running	216 (2m35s ago)	6h42m
kube-system	efs-csi-node-brtwx	3/3	Running	222 (10m ago)	6h42m
kube-system	efs-csi-node-c2t24	0/3	CrashLoopBackOff	189 (59s ago)	6h42m
kube-system	efs-csi-node-r947d	3/3	Running	0	6h42m
kube-system	etcd-k8s-master	1/1	Running	7 (7h37m ago)	10d
kube-system	kube-apiserver-k8s-master	1/1	Running	7 (7h37m ago)	10d
kube-system	kube-controller-manager-k8s-master	1/1	Running	7 (7h37m ago)	10d
kube-system	kube-proxy-9p8lz	1/1	Running	442 (6m21s ago)	10d
kube-system	kube-proxy-k976w	1/1	Running	474 (4m46s ago)	10d
kube-system	kube-proxy-s9nts	1/1	Running	7 (7h37m ago)	10d
kube-system	kube-scheduler-k8s-master	1/1	Running	7 (7h37m ago)	10d
prod-environment	prod-pod	1/1	Running	207 (6m4s ago)	2d3h
staging-environment	staging-pod	1/1	Running	2 (7h37m ago)	2d3h

Phase 3: Backup

Back up application data and the cluster configuration before upgrading. Identify the etcd pod:

```
kubectl get pods -n kube-system | grep etcd
```

```

root@k8s-master:~# kubectl get pods -n kube-system | grep etcd
etcd-k8s-master          1/1      Running    7 (7h40m ago)    10d
root@k8s-master:~#

```

Back up the Kubernetes configuration:

```
sudo cp -a /etc/kubernetes /etc/kubernetes.backup
```

In a production environment, also take an etcd snapshot.

Phase 4: Upgrade the Control Plane to 1.27

Step 1: Configure the 1.27 repository

On the control-plane node, point the package repository to the Kubernetes 1.27 repository on pkgs.k8s.io (the legacy apt.kubernetes.io repository is deprecated). Then update the package index and list the available kubeadm versions:

```

sudo apt-get update
apt-cache madison kubeadm

```

Select a 1.27 patch release, for example 1.27.16-1.1.

```

root@k8s-master:~# sudo apt-get update
Hit:1 http://us-east-1.ec2.archive.ubuntu.com/ubuntu resolute InRelease
Get:2 http://us-east-1.ec2.archive.ubuntu.com/ubuntu resolute-updates InRelease [137 kB]
Get:3 http://us-east-1.ec2.archive.ubuntu.com/ubuntu resolute-backports InRelease [137 kB]
Get:4 http://us-east-1.ec2.archive.ubuntu.com/ubuntu resolute-updates/main amd64v3 Packages [718 kB]
Get:5 http://us-east-1.ec2.archive.ubuntu.com/ubuntu resolute-updates/main Translation-en [167 kB]
Get:6 http://us-east-1.ec2.archive.ubuntu.com/ubuntu resolute-updates/main amd64 Components [97.7 kB]
Get:7 http://us-east-1.ec2.archive.ubuntu.com/ubuntu resolute-updates/universe amd64v3 Packages [316 kB]
Get:8 http://us-east-1.ec2.archive.ubuntu.com/ubuntu resolute-updates/universe Translation-en [101 kB]
Get:9 http://us-east-1.ec2.archive.ubuntu.com/ubuntu resolute-updates/universe amd64 Components [201 kB]
Get:10 http://us-east-1.ec2.archive.ubuntu.com/ubuntu resolute-updates/restricted amd64v3 Packages [487 kB]
Get:11 http://security.ubuntu.com/ubuntu resolute-security InRelease [137 kB]
Get:12 http://us-east-1.ec2.archive.ubuntu.com/ubuntu resolute-updates/restricted Translation-en [97.7 kB]
Get:14 http://us-east-1.ec2.archive.ubuntu.com/ubuntu resolute-updates/multiverse amd64v3 Packages [11.9 kB]
Get:15 http://us-east-1.ec2.archive.ubuntu.com/ubuntu resolute-backports/universe amd64 Components [1056 B]
Hit:13 https://prod-cdn.packages.k8s.io/repositories/isv:/kubernetes:/core:/stable:/v1.29/deb InRelease
Get:16 http://security.ubuntu.com/ubuntu resolute-security/main amd64v3 Packages [551 kB]
Get:17 http://security.ubuntu.com/ubuntu resolute-security/main Translation-en [129 kB]
Get:18 http://security.ubuntu.com/ubuntu resolute-security/main amd64 Components [46.7 kB]
Get:19 http://security.ubuntu.com/ubuntu resolute-security/universe amd64v3 Packages [192 kB]
Get:20 http://security.ubuntu.com/ubuntu resolute-security/universe Translation-en [60.4 kB]
Get:21 http://security.ubuntu.com/ubuntu resolute-security/universe amd64 Components [55.6 kB]
Fetched 3644 kB in 1s (3864 kB/s)
Reading package lists... Done
root@k8s-master:~# sudo apt-get update
Hit:1 http://us-east-1.ec2.archive.ubuntu.com/ubuntu resolute InRelease
Hit:2 http://us-east-1.ec2.archive.ubuntu.com/ubuntu resolute-updates InRelease
Hit:3 http://us-east-1.ec2.archive.ubuntu.com/ubuntu resolute-backports InRelease
Hit:5 http://security.ubuntu.com/ubuntu resolute-security InRelease
Hit:4 https://prod-cdn.packages.k8s.io/repositories/isv:/kubernetes:/core:/stable:/v1.29/deb InRelease
Reading package lists... Done
root@k8s-master:~#

```

```

root@k8s-master:~# apt-cache madison kubeadm
kubeadm | 1.29.15-1.1 | https://pkgs.k8s.io/core:/stable:/v1.29/deb | Packages
kubeadm | 1.29.14-1.1 | https://pkgs.k8s.io/core:/stable:/v1.29/deb | Packages
kubeadm | 1.29.13-1.1 | https://pkgs.k8s.io/core:/stable:/v1.29/deb | Packages
kubeadm | 1.29.12-1.1 | https://pkgs.k8s.io/core:/stable:/v1.29/deb | Packages
kubeadm | 1.29.11-1.1 | https://pkgs.k8s.io/core:/stable:/v1.29/deb | Packages
kubeadm | 1.29.10-1.1 | https://pkgs.k8s.io/core:/stable:/v1.29/deb | Packages
kubeadm | 1.29.9-1.1 | https://pkgs.k8s.io/core:/stable:/v1.29/deb | Packages
kubeadm | 1.29.8-1.1 | https://pkgs.k8s.io/core:/stable:/v1.29/deb | Packages
kubeadm | 1.29.7-1.1 | https://pkgs.k8s.io/core:/stable:/v1.29/deb | Packages
kubeadm | 1.29.6-1.1 | https://pkgs.k8s.io/core:/stable:/v1.29/deb | Packages
kubeadm | 1.29.5-1.1 | https://pkgs.k8s.io/core:/stable:/v1.29/deb | Packages
kubeadm | 1.29.4-2.1 | https://pkgs.k8s.io/core:/stable:/v1.29/deb | Packages
kubeadm | 1.29.3-1.1 | https://pkgs.k8s.io/core:/stable:/v1.29/deb | Packages
kubeadm | 1.29.2-1.1 | https://pkgs.k8s.io/core:/stable:/v1.29/deb | Packages
kubeadm | 1.29.1-1.1 | https://pkgs.k8s.io/core:/stable:/v1.29/deb | Packages
kubeadm | 1.29.0-1.1 | https://pkgs.k8s.io/core:/stable:/v1.29/deb | Packages

```

Step 2: Upgrade kubeadm

```
sudo apt-mark unhold kubeadm
```

```

root@k8s-master:~# sudo apt-mark unhold kubeadm
Canceled hold on kubeadm.
root@k8s-master:~#

```

```
sudo apt-get update && sudo apt-get install -y kubeadm=1.27.16-1.1
```

```

root@k8s-master:~# sudo apt-get update && sudo apt-get install -y kubeadm=1.27.16-1.1
Hit:1 http://us-east-1.ec2.archive.ubuntu.com/ubuntu resolute InRelease
Hit:2 http://us-east-1.ec2.archive.ubuntu.com/ubuntu resolute-updates InRelease
Hit:3 http://us-east-1.ec2.archive.ubuntu.com/ubuntu resolute-backports InRelease
Hit:4 http://security.ubuntu.com/ubuntu resolute-security InRelease
Hit:5 https://prod-cdn.packages.k8s.io/repositories/isv/kubernetes:/core/stable:/v1.27/deb InRelease
Reading package lists... Done
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
Solving dependencies... Done
The following package was automatically installed and is no longer required:
  pollinate
Use 'sudo apt autoremove' to remove it.
The following held packages will be changed:
  kubeadm
The following packages will be DOWNGRADED:
  kubeadm
0 upgraded, 0 newly installed, 1 downgraded, 0 to remove and 18 not upgraded.
E: Packages were downgraded and -y was used without --allow-downgrades.
root@k8s-master:~#

```

```
sudo apt-mark hold kubeadm
```

```

E: Packages were downgraded and -y was used without --allow-downgrades.
root@k8s-master:~# sudo apt-mark hold kubeadm
kubeadm was already set on hold.

```

Step 3: Drain the control-plane node

```
kubectl drain k8s-master --ignore-daemonsets --delete-emptydir-data
```

```

root@k8s-master:~# kubectl drain k8s-master --ignore-daemonsets --delete-emptydir-data
node/k8s-master cordoned
warning: ignoring DaemonSet-managed Pods: kube-system/calico-node-scmjx, kube-system/efs-csi-node-jfswm, kube-system/efs-csi-node-r947d, kube-system/kube-proxy-s9nts
evicting pod kube-system/coredns-76f75df574-m964x
evicting pod kube-system/calico-kube-controllers-8d76c5f9b-c2j8s
pod/calico-kube-controllers-8d76c5f9b-c2j8s evicted
pod/coredns-76f75df574-clw82 evicted
pod/coredns-76f75df574-m964x evicted
node/k8s-master drained

```

Step 4: Verify the kubeadm version

```
kubeadm version
```

Expected: v1.27.x

```

root@k8s-master:~# kubeadm version
kubeadm version: &version.Info{Major:"1", Minor:"27", GitVersion:"v1.27.16", GitCommit:"cbb86e0d7f4a049666fac0551e8b02ef3d6c3d9a", GitTreeState:"clean",
Version:"go1.22.5", Compiler:"gc", Platform:"linux/amd64"}
root@k8s-master:~#

```

Step 5: Check the upgrade plan

```
sudo kubeadm upgrade plan
```

The plan validates that the cluster can be upgraded and should list a 1.27 target version.

Step 6: Apply the control-plane upgrade

```
sudo kubeadm upgrade apply v1.27.16
```

Use the exact patch version shown in the upgrade plan. On success, kubeadm reports that the control-plane upgrade completed.

Phase 5: Upgrade kubelet and kubectl on the Control Plane

```

sudo apt-mark unhold kubelet kubectl
sudo apt-get update
sudo apt-get install -y kubelet=1.27.16-1.1 kubectl=1.27.16-1.1

```

```

root@k8s-master:~# apt-mark unhold kubelet kubect1
kubect1 was already not on hold.
Canceled hold on kubelet.
root@k8s-master:~# sudo apt-get update
Hit:1 http://us-east-1.ec2.archive.ubuntu.com/ubuntu resolute InRelease
Hit:2 http://us-east-1.ec2.archive.ubuntu.com/ubuntu resolute-updates InRelease
Hit:3 http://us-east-1.ec2.archive.ubuntu.com/ubuntu resolute-backports InRelease
Hit:4 http://security.ubuntu.com/ubuntu resolute-security InRelease
Hit:5 https://prod-cdn.packages.k8s.io/repositories/isv:/kubernetes:/core:/stable:/v1.27/deb InRelease
Reading package lists... Done
root@k8s-master:~# sudo apt-get install -y kubelet=1.27.16-1.1 kubect1=1.27.16-1.1
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
Solving dependencies... Done
The following package was automatically installed and is no longer required:
  pollinate
Use 'sudo apt autoremove' to remove it.
The following additional packages will be installed:
  ebttables socat
The following NEW packages will be installed:
  ebttables socat
The following packages will be DOWNGRADED:
  kubect1 kubelet
0 upgraded, 2 newly installed, 2 downgraded, 0 to remove and 18 not upgraded.
E: Packages were downgraded and -y was used without --allow-downgrades.
root@k8s-master:~#

```

```

sudo apt-mark hold kubelet kubect1
sudo systemctl daemon-reload
sudo systemctl restart kubelet
sudo systemctl status kubelet

```

Expected: Active: active (running)

```

root@k8s-master:~# sudo apt-mark hold kubelet kubect1
kubelet set on hold.
kubect1 set on hold.
root@k8s-master:~# sudo systemctl daemon-reload
root@k8s-master:~# sudo systemctl restart kubelet
root@k8s-master:~# sudo systemctl status kubelet
● kubelet.service - kubelet: The Kubernetes Node Agent
   Loaded: loaded (/usr/lib/systemd/system/kubelet.service; enabled; preset: enabled)
  Drop-In: /usr/lib/systemd/system/kubelet.service.d
           └─10-kubeadm.conf
   Active: active (running) since Fri 2026-10-02 15:07:17 UTC; 20s ago
     Invocation: b4bad8b65aba4ecd9138d41e2e4e7b8f
       Docs: https://kubernetes.io/docs/
    Main PID: 367264 (kubelet)
       Tasks: 10 (limit: 3712)
     Memory: 31.7M (peak: 32M)
        CPU: 572ms
     CGroup: /system.slice/kubelet.service
             └─367264 /usr/bin/kubelet --bootstrap-kubeconfig=/etc/kubernetes/bootstrap-kubelet.conf --kubeconfig=/etc/kubernetes/kubelet.conf --config=/var/lib/kubelet/co

```

```

Oct 02 15:07:19 k8s-master kubelet[367264]: I1002 15:07:19.052160 367264 reconciler_common.go:258] "operationExecutor.VerifyControllerAttachedVolume started for volume "
Oct 02 15:07:19 k8s-master kubelet[367264]: I1002 15:07:19.052175 367264 reconciler_common.go:258] "operationExecutor.VerifyControllerAttachedVolume started for volume "
Oct 02 15:07:19 k8s-master kubelet[367264]: I1002 15:07:19.052189 367264 reconciler_common.go:258] "operationExecutor.VerifyControllerAttachedVolume started for volume "
Oct 02 15:07:19 k8s-master kubelet[367264]: I1002 15:07:19.052343 367264 reconciler_common.go:258] "operationExecutor.VerifyControllerAttachedVolume started for volume "
Oct 02 15:07:19 k8s-master kubelet[367264]: I1002 15:07:19.052399 367264 reconciler_common.go:258] "operationExecutor.VerifyControllerAttachedVolume started for volume "
Oct 02 15:07:19 k8s-master kubelet[367264]: I1002 15:07:19.052418 367264 reconciler_common.go:258] "operationExecutor.VerifyControllerAttachedVolume started for volume "
Oct 02 15:07:19 k8s-master kubelet[367264]: I1002 15:07:19.052676 367264 reconciler_common.go:258] "operationExecutor.VerifyControllerAttachedVolume started for volume "
Oct 02 15:07:19 k8s-master kubelet[367264]: I1002 15:07:19.052714 367264 reconciler_common.go:258] "operationExecutor.VerifyControllerAttachedVolume started for volume "
Oct 02 15:07:19 k8s-master kubelet[367264]: I1002 15:07:19.052753 367264 reconciler_common.go:258] "operationExecutor.VerifyControllerAttachedVolume started for volume "
Oct 02 15:07:19 k8s-master kubelet[367264]: I1002 15:07:19.052788 367264 reconciler_common.go:258] "operationExecutor.VerifyControllerAttachedVolume started for volume "
lines 1-24/24 (END)

```

Phase 6: Verify the 1.27 Control Plane

```
kubect1 get nodes
```

The control plane is at v1.27.x while the workers are still at v1.26.x. This is expected during a rolling upgrade.

```

root@k8s-master:~# kubect1 get nodes
NAME             STATUS              ROLES    AGE   VERSION
k8s-master       Ready               control-plane   35m   v1.27.16
k8s-worker1      Ready,SchedulingDisabled <none>    28m   v1.26.15
k8s-worker2      Ready               <none>    28m   v1.26.15
root@k8s-master:~#

```

Phase 7: Drain Worker 1

Run on the control plane:

```
kubectl drain k8s-worker1 --ignore-daemonsets --delete-emptydir-data
kubectl get nodes
```

Expected: k8s-worker1 shows Ready,SchedulingDisabled.

```
root@k8s-master:~# kubectl drain k8s-worker1 --ignore-daemonsets --delete-emptydir-data
node/k8s-worker1 already cordoned
Warning: ignoring DaemonSet-managed Pods: kube-system/calico-node-4d7hl, kube-system/kube-proxy-l88bf
node/k8s-worker1 drained
root@k8s-master:~# kubectl get nodes
```

NAME	STATUS	ROLES	AGE	VERSION
k8s-master	Ready	control-plane	37m	v1.27.16
k8s-worker1	Ready,SchedulingDisabled	<none>	30m	v1.26.15
k8s-worker2	Ready	<none>	30m	v1.26.15

```
root@k8s-master:~#
```

Phase 8: Upgrade kubeadm on Worker 1

SSH to worker 1 and check the current version:

```
ssh ubuntu@<worker1-ip>
kubeadm version
```

Configure the 1.27 repository on the worker, then run:

```
sudo apt-mark unhold kubeadm
sudo apt-get update
```

```
root@k8s-worker1:~# kubeadm version
kubeadm version: &version.Info{Major:"1", Minor:"26", GitVersion:"v1.26.15", GitCommit:"1649f592f1909b97aa3c2a0a8f968a3fd05a7b8b", GitTreeState:"clean", BuildDate:"2024-03-01T18:08:19Z", GoVersion:"go1.22.3", Compiler:"gc", Platform:"linux/amd64"}
root@k8s-worker1:~# sudo apt-get update
Hit:1 http://us-east-1.ec2.archive.ubuntu.com/ubuntu resolute InRelease
Hit:2 http://us-east-1.ec2.archive.ubuntu.com/ubuntu resolute-updates InRelease
Hit:3 http://us-east-1.ec2.archive.ubuntu.com/ubuntu resolute-backports InRelease
Hit:4 http://security.ubuntu.com/ubuntu resolute-security InRelease
Hit:5 https://download.docker.com/linux/ubuntu resolute InRelease
Hit:6 https://prod-cdn.packages.k8s.io/repositories/ipv:/kubernetes:/core:/stable:/v1.26/deb InRelease
Reading package lists... Done
```

```
root@k8s-worker1:~# sudo apt-mark unhold kubeadm
Canceled hold on kubeadm.
root@k8s-worker1:~#
```

```
sudo apt-get install -y kubeadm=1.27.16-1.1
sudo apt-mark hold kubeadm
sudo kubeadm upgrade node
```

```
root@k8s-worker1:~# sudo apt-get install -y kubeadm=1.27.16-1.1
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
Package kubeadm is not available, but is referred to by another package.
This may mean that the package is missing, has been obsoleted, or
is only available from another source.

E: Version '1.27.16-1.1' for 'kubeadm' was not found
root@k8s-worker1:~# sudo apt-mark hold kubeadm
kubeadm set on hold.
root@k8s-worker1:~# sudo kubeadm upgrade node
[upgrade] Reading configuration from the cluster...
[upgrade] FYI: You can look at this config file with 'kubectl -n kube-system get cm kubeadm-config -o yaml'
W1006 11:34:22.047148 72232 configset.go:177] error unmarshaling configuration schema.GroupVersionKind{Group:"kubernetes.k8s.io", Version:"v1beta1", Kind:"KubeletConfiguration"}: strict decoding error: unknown field "containerRuntimeEndpoint"
[preflight] Running pre-flight checks
[preflight] Skipping pre-pull. Not a control plane node.
[upgrade] Skipping phase. Not a control plane node.
[kubelet-start] Writing kubelet configuration to file "/var/lib/kubelet/config.yaml"
[upgrade] The configuration for this node was successfully updated!
[upgrade] Now you should go ahead and upgrade the kubelet package using your package manager.
root@k8s-worker1:~#
```

Phase 9: Upgrade kubelet and kubectl on Worker 1

```
sudo apt-mark unhold kubelet kubectl
sudo apt-get update
```

```

root@k8s-worker1:~# sudo apt-mark unhold kubelet kubect1
Canceled hold on kubelet.
Canceled hold on kubect1.
root@k8s-worker1:~# sudo apt-get update
Hit:1 http://us-east-1.ec2.archive.ubuntu.com/ubuntu resolute InRelease
Hit:2 https://download.docker.com/linux/ubuntu resolute InRelease
Hit:3 http://us-east-1.ec2.archive.ubuntu.com/ubuntu resolute-updates InRelease
Hit:4 http://us-east-1.ec2.archive.ubuntu.com/ubuntu resolute-backports InRelease
Hit:5 https://prod-cdn.packages.k8s.io/repositories/isv:/kubernetes:/core:/stable:/v1.26/deb InRelease
Hit:6 http://security.ubuntu.com/ubuntu resolute-security InRelease
Reading package lists... Done

```

```
sudo apt-get install -y kubelet=1.27.16-1.1 kubect1=1.27.16-1.1
```

```

root@k8s-worker1:~# sudo apt-get install -y kubelet=1.27.16-1.1 kubect1=1.27.16-1.1
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
Solving dependencies... Done
The following package was automatically installed and is no longer required:
  pollinate
Use 'sudo apt autoremove' to remove it.
The following packages will be upgraded:
  kubect1 kubelet
2 upgraded, 0 newly installed, 0 to remove and 22 not upgraded.
Need to get 29.5 MB of archives.
After this operation, 15.3 MB disk space will be freed.
Get:1 https://prod-cdn.packages.k8s.io/repositories/isv:/kubernetes:/core:/stable:/v1.27/deb kubect1 1.27.16-1.1 [10.4 MB]
Get:2 https://prod-cdn.packages.k8s.io/repositories/isv:/kubernetes:/core:/stable:/v1.27/deb kubelet 1.27.16-1.1 [19.2 MB]
Fetched 29.5 MB in 0s (116 MB/s)
(Reading database ... 124363 files and directories currently installed.)
Preparing to unpack .../kubect1_1.27.16-1.1_amd64.deb ...
Unpacking kubect1 (1.27.16-1.1) over (1.26.15-1.1) ...
Preparing to unpack .../kubelet_1.27.16-1.1_amd64.deb ...
Unpacking kubelet (1.27.16-1.1) over (1.26.15-1.1) ...
dpkg: warning: unable to delete old directory '/etc/sysconfig': Directory not empty
Setting up kubect1 (1.27.16-1.1) ...
Setting up kubelet (1.27.16-1.1) ...
Scanning processes...
Scanning candidates...
Scanning linux images...

Running kernel seems to be up-to-date.

Restarting services...
systemctl restart kubelet.service

No containers need to be restarted.

No user sessions are running outdated binaries.

No VM guests are running outdated hypervisor (qemu) binaries on this host.
root@k8s-worker1:~#

```

```

sudo apt-mark hold kubelet kubect1
sudo systemctl daemon-reload
sudo systemctl restart kubelet
kubelet --version

```

```

root@k8s-worker1:~# sudo apt-mark hold kubelet kubect1
kubelet set on hold.
kubect1 set on hold.
root@k8s-worker1:~# sudo systemctl daemon-reload
root@k8s-worker1:~#
^C
root@k8s-worker1:~#
^C
root@k8s-worker1:~# sudo systemctl restart kubelet
root@k8s-worker1:~# kubelet --version
Kubernetes v1.27.16
root@k8s-worker1:~#

```

Phase 10: Uncordon Worker 1

Run on the control plane:

```

kubect1 uncordon k8s-worker1
kubect1 get nodes

```

Expected: k8s-master and k8s-worker1 at v1.27.x; k8s-worker2 still at v1.26.x.

```

root@k8s-master:~# kubectl uncordon k8s-worker1
node/k8s-worker1 uncordoned
root@k8s-master:~# kubectl get nodes
NAME                STATUS    ROLES    AGE   VERSION
k8s-master          Ready     control-plane   56m   v1.27.16
k8s-worker1         Ready     <none>        49m   v1.27.16
k8s-worker2         Ready     <none>        49m   v1.26.15
root@k8s-master:~#

```

Phase 11: Upgrade Worker 2

Repeat the same process as for worker 1. Drain the node from the control plane:

```
kubectl drain k8s-worker2 --ignore-daemonsets --delete-emptydir-data
```

SSH to worker 2 and upgrade kubeadm:

```

ssh ubuntu@<worker2-ip>
sudo apt-mark unhold kubeadm
sudo apt-get update
sudo apt-get install -y kubeadm=1.27.16-1.1

```

```

root@k8s-worker2:~# sudo apt-mark unhold kubeadm
Canceled hold on kubeadm.
root@k8s-worker2:~# sudo apt-get update
Hit:1 http://us-east-1.ec2.archive.ubuntu.com/ubuntu resolute InRelease
Hit:2 http://us-east-1.ec2.archive.ubuntu.com/ubuntu resolute-updates InRelease
Hit:3 http://us-east-1.ec2.archive.ubuntu.com/ubuntu resolute-backports InRelease
Get:5 http://security.ubuntu.com/ubuntu resolute-security InRelease [137 kB]
Hit:4 https://prod-cdn.packages.k8s.io/repositories/iscv/kubernetes:/core:/stable:/v1.26/deb InRelease
Get:6 http://security.ubuntu.com/ubuntu resolute-security/main amd64v3 Packages [563 kB]
Get:7 http://security.ubuntu.com/ubuntu resolute-security/main Translation-en [132 kB]
Fetched 832 kB in 1s (921 kB/s)
Reading package lists... Done
root@k8s-worker2:~# sudo apt-get install -y kubeadm=1.27.16-1.1
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
Package kubeadm is not available, but is referred to by another package.
This may mean that the package is missing, has been obsoleted, or
is only available from another source

E: Version '1.27.16-1.1' for 'kubeadm' was not found

```

```

sudo apt-mark hold kubeadm
sudo kubeadm upgrade node

```

Upgrade kubelet and kubectl:

```

sudo apt-mark unhold kubelet kubectl
sudo apt-get update

```

```

root@k8s-worker2:~# sudo kubeadm upgrade node
[upgrade] Reading configuration from the cluster...
[upgrade] FYI: You can look at this config file with 'kubectl -n kube-system get cm kubeadm-config -o yaml'
W006 11:48:15.388863 119797 configset.go:177] error unmarshaling configuration schema.GroupVersionKind(group:"kubelet.config.k8s.io", version:"v1beta1", kind:"KubeletConfiguration"): strict decoding error: unknown field "containerRuntimeEndpoint"
[preflight] Running pre-flight checks
[preflight] Skipping preflight. Not a control plane node.
[upgrade] Skipping phase. Not a control plane node.
[kubelet-start] Writing kubelet configuration to file "/var/lib/kubelet/config.yaml"
[upgrade] The configuration for this node was successfully updated!
[upgrade] Now you should go ahead and upgrade the kubelet package using your package manager.
root@k8s-worker2:~# sudo apt-mark unhold kubelet kubectl
Canceled hold on kubelet.
root@k8s-worker2:~# sudo apt-get update
Hit:1 http://us-east-1.ec2.archive.ubuntu.com/ubuntu resolute InRelease
Hit:2 http://us-east-1.ec2.archive.ubuntu.com/ubuntu resolute-updates InRelease
Hit:3 http://us-east-1.ec2.archive.ubuntu.com/ubuntu resolute-backports InRelease
Hit:4 http://security.ubuntu.com/ubuntu resolute-security InRelease
Hit:5 https://prod-cdn.packages.k8s.io/repositories/iscv/kubernetes:/core:/stable:/v1.27/deb InRelease
Reading package lists... Done
root@k8s-worker2:~#

```

```
sudo apt-get install -y kubelet=1.27.16-1.1 kubectl=1.27.16-1.1
```

```

root@k8s-worker2:~# sudo apt-get install -y kubelet=1.27.16-1.1 kubectl=1.27.16-1.1
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
Solving dependencies... Done
The following package was automatically installed and is no longer required:
  pollinate
Use 'sudo apt autoremove' to remove it.
The following packages will be upgraded:
  kubectl kubelet
2 upgraded, 0 newly installed, 0 to remove and 19 not upgraded.
Need to get 29.5 MB of archives.
After this operation, 15.3 MB disk space will be freed.
Get:1 https://prod-cdn.packages.k8s.io/repositories/isyv:/kubernetes/core/stable/v1.27/deb kubectl 1.27.16-1.1 [10.4 MB]
Get:2 https://prod-cdn.packages.k8s.io/repositories/isyv:/kubernetes/core/stable/v1.27/deb kubelet 1.27.16-1.1 [19.2 MB]
Fetched 29.5 MB in 0s (136 MB/s)
(Reading database ... 124494 files and directories currently installed.)
Preparing to unpack .../kubectl_1.27.16-1.1_amd64.deb ...
Unpacking kubectl (1.27.16-1.1) over (1.26.15-1.1) ...
Preparing to unpack .../kubelet_1.27.16-1.1_amd64.deb ...
Unpacking kubelet (1.27.16-1.1) over (1.26.15-1.1) ...
dpkg: warning: unable to delete old directory '/etc/sysconfig': Directory not empty
Setting up kubectl (1.27.16-1.1) ...
Setting up kubelet (1.27.16-1.1) ...
Scanning processes...
Scanning candidates...
Scanning linux images...

Running kernel seems to be up-to-date.

Restarting services...
systemctl restart kubelet.service

Service restarts being deferred:
systemctl restart networkd-dispatcher.service
systemctl restart systemd-logind.service
systemctl restart unattended-upgrades.service

No containers need to be restarted.

No user sessions are running outdated binaries.

No VM guests are running outdated hypervisor (qemu) binaries on this host.

```

```

sudo apt-mark hold kubelet kubectl
sudo systemctl daemon-reload
sudo systemctl restart kubelet
sudo systemctl status kubelet --no-pager

```

Expected: Active: active (running)

```

root@k8s-worker2:~# sudo apt-mark hold kubelet kubectl
kubelet set on hold.
kubectl set on hold.
root@k8s-worker2:~# sudo systemctl daemon-reload
sudo systemctl restart kubelet
root@k8s-worker2:~# sudo systemctl status kubelet --no-pager
● kubelet.service - kubelet: The Kubernetes Node Agent
   Loaded: loaded (/usr/lib/systemd/system/kubelet.service; enabled; preset: enabled)
   Drop-In: /usr/lib/systemd/system/kubelet.service.d
            -10-kubeadm.conf
   Active: active (running) since Tue 2026-10-06 11:50:34 UTC; 10s ago
   Invocation: 5b7b42c57e6041639a543748582267f
   Docs: https://kubernetes.io/docs/
   Main PID: 121972 (kubelet)
   Tasks: 10 (limit: 3712)
   Memory: 25.4M (peak: 25.4M)
   CPU: 272ms
   CGroup: /system.slice/kubelet.service
            └─121972 /usr/bin/kubelet --bootstrap-kubeconfig=/etc/kubernetes/bootstrap-kubelet.conf --kubeconfig=/etc/kubernetes/kubelet.conf --config=/var/lib/kubelet/config.yaml --container-runtime-endpoint=
Oct 06 11:50:38 k8s-worker2 kubelet[121972]: I1006 11:50:38.480275 121972 scope.go:115] "RemoveContainer" containerID="b66f47ea231f0b64a94c17ab734fc15943dc85fe33c4daff56868907f8f436c8"
Oct 06 11:50:38 k8s-worker2 kubelet[121972]: I1006 11:50:38.482616 121972 scope.go:115] "RemoveContainer" containerID="570f869da0f3998f645d11cee6ef659e1856c24b3016688e08783a88d4ef65e"
Oct 06 11:50:38 k8s-worker2 kubelet[121972]: I1006 11:50:38.494320 121972 scope.go:115] "RemoveContainer" containerID="cb40813468a00fd4ec22a0d20b1bbf6f8f66fc3fe24398f49ab11f42c4fa858"
Oct 06 11:50:38 k8s-worker2 kubelet[121972]: I1006 11:50:38.501303 121972 scope.go:115] "RemoveContainer" containerID="847701d60dbb8e43745afe08f70518bbef70b30524d704108df9bd10bd1ebbc"
Oct 06 11:50:41 k8s-worker2 kubelet[121972]: I1006 11:50:41.496451 121972 scope.go:115] "RemoveContainer" containerID="460b0834a997aa0105aec91126b6c9193c15eb58107efc1489da18831ea75a5a"
Oct 06 11:50:41 k8s-worker2 kubelet[121972]: E1006 11:50:41.496802 121972 pod_workers.go:1294] "Error syncing pod, skipping" err="failed to 'StartContainer' for 'calico-node' with CrashLoopBackOff: 'back"
Oct 06 11:50:42 k8s-worker2 kubelet[121972]: I1006 11:50:42.497624 121972 scope.go:115] "RemoveContainer" containerID="460b0834a997aa0105aec91126b6c9193c15eb58107efc1489da18831ea75a5a"
Oct 06 11:50:42 k8s-worker2 kubelet[121972]: E1006 11:50:42.497950 121972 pod_workers.go:1294] "Error syncing pod, skipping" err="failed to 'StartContainer' for 'calico-node' with CrashLoopBackOff: 'back"
Oct 06 11:50:43 k8s-worker2 kubelet[121972]: I1006 11:50:43.924396 121972 scope.go:115] "RemoveContainer" containerID="460b0834a997aa0105aec91126b6c9193c15eb58107efc1489da18831ea75a5a"
Oct 06 11:50:43 k8s-worker2 kubelet[121972]: E1006 11:50:43.924761 121972 pod_workers.go:1294] "Error syncing pod, skipping" err="failed to 'StartContainer' for 'calico-node' with CrashLoopBackOff: 'back"
Hint: Some lines were ellipsized, use -l to show in full.
root@k8s-worker2:~#

```

Uncordon the node from the control plane and verify:

```

kubectl uncordon k8s-worker2
kubectl get nodes

```

Expected: all nodes at v1.27.x.

```

root@k8s-master:~# kubectl get nodes
NAME          STATUS    ROLES          AGE    VERSION
k8s-master   Ready     control-plane   64m    v1.27.16
k8s-worker1   Ready     <none>          57m    v1.27.16
k8s-worker2   Ready     <none>          57m    v1.27.16
root@k8s-master:~#

```

Phase 12: Upgrade from 1.27 to 1.28

Repeat the same process using the 1.28 repository and 1.28.x packages.

Control plane:

1. Upgrade kubeadm
2. `sudo kubeadm upgrade plan`
3. `sudo kubeadm upgrade apply v1.28.x`
4. Upgrade kubelet
5. Upgrade kubectl

Workers (one at a time):

1. Drain the node
2. Upgrade kubeadm
3. `sudo kubeadm upgrade node`
4. Upgrade kubelet
5. Upgrade kubectl
6. Restart kubelet
7. Uncordon the node

Upgrade Sequence Summary

Stage	Node	Actions
Pre-checks	Control plane	Check cluster health; back up /etc/kubernetes and etcd
1.26 to 1.27	Control plane	kubeadm, upgrade plan, upgrade apply, kubelet, kubectl
1.26 to 1.27	Worker 1	drain, kubeadm, upgrade node, kubelet, kubectl, restart kubelet, uncordon
1.26 to 1.27	Worker 2	Same steps as worker 1
Verify	All nodes	All nodes report v1.27.x
1.27 to 1.28	Control plane, then workers	Same sequence as above, one node at a time
Final state	All nodes	Cluster at v1.28.x